FastSpeech 2 Text-to-Speech for English and Telugu

Technical Report

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Date: 7 November 2026

A project report describing the implementation, training and evaluation of two FastSpeech 2 speech-synthesis systems.

Abstract

This project implements two FastSpeech 2 text-to-speech systems: one for Indian-accented English using the Svarah corpus and another for Telugu using the IndicTTS Telugu corpus. Both models were trained from scratch using approximately five hours of selected speech per language. The work covers data preparation, text normalisation, phoneme conversion, forced alignment, acoustic-model training and speech generation.

The English pipeline uses an existing pronunciation lexicon together with g2p_en, while the Telugu pipeline required a rule-based text front-end because the reference implementation did not include a Telugu configuration. This front-end maps Telugu script to a set of 47 phones and expands numbers and time expressions into words. Corpus-specific Montreal Forced Aligner models were trained to obtain phoneme durations. A pause-aware alignment adjustment was also introduced for English.

The final models generated intelligible speech for all ten assignment test sentences. Listening observations and per-phoneme duration, pitch and energy predictions were used to examine how the systems handled long or unfamiliar words, heteronyms, Telugu homographs, numerals and punctuation. The results also revealed areas that need further work, particularly rare phoneme sequences, Telugu number handling and pause modelling. The listening assessment was informal and involved a single listener, so the results should be treated as an initial evaluation rather than a formal perceptual study.

1. Introduction and project overview

The aim of this project was to build text-to-speech systems for two languages using the FastSpeech 2 architecture: Indian-accented English and Telugu. The implementation is based on the open-source ming024/FastSpeech2 repository [7], with changes to support the selected datasets and the Telugu phoneme inventory.

The work was organised as two language-specific pipelines. Each pipeline includes audio and transcript selection, text cleaning, grapheme-to-phoneme conversion, forced alignment, acoustic-feature extraction, model training and inference. The English model uses Svarah, while the Telugu model uses IndicTTS Telugu.

A key part of the project was adapting the text-processing stage for Telugu. Since the reference repository did not provide a Telugu recipe, a rule-based converter was developed to map Telugu script to 47 phones. A Telugu number reader was also added so that numerals, ordinals and time expressions could be spoken as words.

The project also examines the variance adaptor, which predicts duration, pitch and energy. These predictions were saved for each test sentence and compared with the listening observations. This helped identify which parts of the input were handled well and which still caused pronunciation or clarity issues.

1.1 Project summary

Measure	English	Telugu
Corpus	Svarah	IndicTTS Telugu
Speakers used	60	2 (TEF and TEM)
Aligned speech used (train + validation)	5.66 hours	4.23 hours
Acoustic-model parameters	35.43 million	35.35 million
Final training step	15,000	22,000
Best validation total loss	2.639 at step 9,000	3.768 at step 1,000

1.2 Main observations

- The English front-end produced different stress patterns for “conduct” when it was used as a noun and as a verb. The variance adaptor reflected this difference mainly through vowel duration and energy.
- Both Telugu test sentences containing □□□□□ were judged very clear, even though the word was mapped to the same phone sequence in both contexts.
- The comma in English test sentence E4 was represented by a pause of approximately 0.30 seconds.
- Telugu test sentence T4 was the most difficult input. Six vowels had predicted energy below 20, and five of those were below 10.

2. System architecture

FastSpeech 2 is a non-autoregressive text-to-speech model. In this project, the input text is first converted into a sequence of phonemes. The Transformer encoder converts that sequence into hidden representations. The variance adaptor then predicts duration, pitch and energy. Duration predictions are used by the length regulator to expand the phoneme-level representations into a frame-level sequence. Pitch and energy information are added before the Transformer decoder generates an 80-bin mel-spectrogram.

A pretrained universal HiFi-GAN vocoder converts the predicted mel-spectrogram into an audio waveform [2]. During training, the model uses the ground-truth duration, pitch and energy targets. During inference, it uses the values predicted by the variance adaptor. Pitch features were extracted using WORLD [4], and energy was calculated as the L2 norm of each STFT frame.

Component	Configuration
Encoder / decoder	4 / 6 Transformer layers (repository configuration)
Hidden size / attention heads	256 / 2
Variance predictors	Conv1D, kernel size 3, 256 filters, dropout 0.5
Pitch and energy	Per phoneme; 256 bins; normalised using training-set statistics
Mel-spectrogram	80 bins; 22.05 kHz; window 1,024; hop 256
Frame duration	Approximately 11.6 ms
Maximum sequence length	1,500
Speaker embeddings	60 English speakers; 2 Telugu speakers
Vocoder	Pretrained universal HiFi-GAN, unchanged
Model parameters	35.43 M (English); 35.35 M (Telugu)

2.1 Changes made to the reference implementation

- Separate utterance-level training and validation lists were added. Normalisation statistics for pitch and energy were calculated from the training data only.
- For English, gaps of sufficient length in the forced-alignment output were converted into explicit pause tokens.
- The original ARPAbet and pinyin symbol tables were replaced with a Telugu phone inventory containing 47 phones and a shared pause token.
- Compatibility changes were made for the PyTorch and librosa versions available in the current runtime. Feature extraction was also made usable on CPU.
- The configuration was adjusted for multi-speaker training, the universal HiFi-GAN vocoder and a maximum sequence length of 1,500.
- An inference script was added to normalise text, convert it to phonemes, generate speech and save duration, pitch, energy and mel-spectrogram plots, together with per-phoneme CSV files.

A copy-synthesis check was also carried out by passing mel-spectrograms extracted from real recordings through the vocoder. The reconstructed audio was close to the original recordings, suggesting that the vocoder was not the main source of the remaining quality issues.

2.2 Differences from the original FastSpeech 2 setup

Aspect	Original paper	This project
Training data	LJSpeech, about 24 hours, one speaker	Approximately 4–6 hours per language, multiple speakers
Decoder depth	4 Transformer blocks	6 blocks, following repository defaults
Pitch and energy	Frame-level; pitch uses continuous wavelet transform	Per-phoneme prediction; pitch regressed directly
Vocoder	Parallel WaveGAN	Pretrained universal HiFi-GAN
Alignment	MFA trained on training data	MFA trained separately for each selected corpus
G2P	Kyubyong g2p, 76 symbols	g2p_en for English; custom 47-phone rules for Telugu
Training	About 160,000 steps, batch size 48	15,000 English and 22,000 Telugu steps, batch size 16
Evaluation	MOS and CMOS listening tests	Listening assessment, validation loss and per-phoneme analysis

3. Data collection and preprocessing

The two corpora were prepared separately, but the same general checks were applied to both. Audio clips were filtered by duration and quality, training and validation data were separated to reduce leakage, and audio was resampled to 22.05 kHz. Each utterance was also peak-normalised.

Preparation detail	English (Svarah)	Telugu (IndicTTS Telugu)
Original corpus	9.6 h; 6,656 clips; 16 kHz	36.7 h; 8,576 clips; 2 speakers
Selection	1–15 second clips; 60 speakers with the most speech; 6.32 h / 4,048 clips	2–15 second clips; characters-per-second outlier filtering
Selected split	5.32 h train (3,380 clips); 1.00 h validation (668 clips)	5.00 h train (2,147 clips); 1.00 h validation (419 clips)
Leakage control	Same sentence or recording kept in one split; validation capped at 25% of each speaker's audio	Matching sentences kept in one split, including sentences recorded by both speakers
Split checks	No sentence, recording or clip overlap; each validation speaker also appears in training	No text or clip overlap
After alignment	3,323 train and 657 validation utterances; 5.66 h aligned	2,136 train and 417 validation utterances; 4.23 h aligned
Resampling	16 kHz to 22.05 kHz	16 kHz and 48 kHz to 22.05 kHz

3.1 English text processing and speaker selection

The English transcripts were converted to lowercase. Numbers, ordinals, percentages, decimals and years were expanded into words, and apostrophes at the edges of tokens were removed. The pronunciation dictionary combined the LibriSpeech lexicon, which contains 206,508 entries, with g2p_en pronunciations for 1,113 words that were not present in that lexicon. Letter-by-letter pronunciations were also added for acronyms such as UPI, OTP and SBI.

The Svarah corpus does not provide explicit speaker IDs. Speaker groups were therefore approximated using available recording metadata, including gender, age group, language, state, district, qualification and occupation. This produced 115 profiles. The 60 profiles with the largest amount of speech were selected for the project.

3.2 Telugu text processing

The Telugu transcripts were cleaned by removing byte-order marks and punctuation. In some corpus clips, the Latin character “o” had been used to represent anusvara. The 46 occurrences found during preprocessing were changed . A rule-based dictionary containing 12,127 word types was then created from the cleaned transcripts. No unhandled characters remained in the resulting text.

3.3 Forced alignment and pause handling

Phoneme durations were obtained using Montreal Forced Aligner (MFA) version 3.4.3 [3]. Separate aligners were trained from scratch for English and Telugu because the available pretrained models did not match the Indian-accented English data or the Telugu phone set. The English aligner used 30 monophone and 25 triphone iterations. The Telugu aligner used the default 40 monophone iterations.

MFA 3 does not include silence intervals in its TextGrid output. The reference preprocessing code, however, calculates durations from labelled intervals only. To check the effect of this behaviour, 300 English utterances were inspected; 258 contained gaps between labelled intervals. In the English pipeline, gaps of at least 30 ms were labelled with the pause token sp. Shorter gaps were added to the preceding phone.

This adjustment added 1.23 hours of pause time and increased the processed English audio from 4.48 to 5.66 hours. As a result, the English duration, pitch and energy targets account for pauses in the audio. The Telugu features were extracted using the standard alignments, without this pause-aware adjustment. Applying the same method to Telugu is therefore included as a future improvement.

3.4 Text-to-phoneme conversion

English: Text normalisation is performed before grapheme-to-phoneme conversion. Numbers, dates, times and a.m./p.m. expressions are expanded into words. The complete sentence is passed to g2p_en so that its part-of-speech tagger can help distinguish heteronyms. The output uses ARPAbet symbols with stress markers, and punctuation is mapped to the pause token sp.

Telugu: A rule-based converter maps Unicode Telugu characters to 47 phones. The rules cover the inherent vowel, virama-based conjuncts, anusvara assimilation according to the nasal class of the following consonant, and geminates represented by repeated consonant phones. A separate number reader expands numerals, ordinals and times. The repository's original symbol table was replaced with the Telugu phone set.

4. Model training and loss analysis

Both models were trained with the Adam optimiser using $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\varepsilon = 1e\text{-}9$. The training setup retained the repository's warm-up learning-rate schedule with 4,000 warm-up steps and gradient clipping. A batch size of 16 was used, and training ran on a Tesla T4 GPU through Google Colab. Checkpoints and log files were saved to Google Drive so that training could be resumed if a Colab session ended.

The English model was trained for 15,000 steps, corresponding to approximately 72 epochs and 2.5 hours of training. Validation was run every 1,500 steps. The Telugu model was trained for 22,000 steps, or approximately 160 epochs, within the configured limit of 30,000 steps. Telugu validation was run every 1,000 steps.

The following values were recorded at the final training step. Training losses are taken from individual logged batches, so small fluctuations are expected. The loss values should not be compared directly across languages because the feature statistics differ.

Loss component	English train	English validation	Telugu train	Telugu validation
Total	1.006	2.659	0.813	4.005

Mel	0.458	0.750	0.376	1.002
Pitch	0.023	0.496	0.028	1.038
Energy	0.044	0.417	0.025	0.889
Duration	0.023	0.247	0.008	0.074

The best validation total loss was 2.639 for English at step 9,000 and 3.768 for Telugu at step 1,000.

4.1 Training behaviour

The training curves decreased steadily in both experiments, with no major spikes or signs of divergence. Total training loss fell from approximately 6 to 1.0 for English and from approximately 7 to 0.8 for Telugu. Most of the reduction came from the mel and post-net losses. Pitch, energy and duration losses settled in the range of 0.02–0.04, with Telugu duration loss reaching 0.008.

English validation loss reached its lowest value, 2.639, at step 9,000. It then remained within approximately 0.03 of that value through step 15,000. Telugu validation loss was lowest at step 1,000, at 3.768, and gradually increased to 4.005 by step 22,000. The increase was mainly in pitch and energy loss; mel and duration validation losses remained almost unchanged.

4.2 Generalisation

The training and validation losses show a noticeable gap in the variance predictors. English pitch loss on validation data was about twenty times the training loss, while the corresponding ratio for Telugu was about thirty-seven. This suggests that pitch prediction is a useful area to investigate when improving generalisation. It is consistent with the importance given to pitch modelling in the FastSpeech 2 paper, but no ablation experiment was carried out here. The interpretation should therefore be treated as an observation from the loss curves, not as a confirmed cause.

4.3 Checkpoint selection

Validation loss measures frame-level prediction error on held-out recordings when ground-truth durations are supplied. It does not directly measure whether the generated sentence is easy to understand or sounds natural. For this reason, the final checkpoints were also compared by listening to the generated test sentences. The later checkpoints were judged clearer, so step 15,000 was selected for English (speaker S000) and step 22,000 for Telugu (speaker TEM).

5. Generated speech and listening assessment

The ten assignment test sentences were synthesised using the selected checkpoints: English step 15,000 with speaker S000 and Telugu step 22,000 with speaker TEM. The generated audio was then reviewed by listening. The Telugu recordings from speaker TEM were clearer than those from TEF, whose recordings contained more noise, so TEM was used for the final Telugu outputs. Phone counts in the following table include pause tokens. Telugu examples are described through English glosses; the exact Telugu inputs are stored in test_sentences_telugu.json and Results/Telugu/phoneme_sequences.txt.

ID	Test input / description	What it tests	Phones	Audio (s)	Phones/s
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Long word not in the lexicon	87	5.14	16.9
E2a	The orchestra's conduct during the ceremony was praised by the audience.	"Conduct" as a noun	51	3.12	16.3
E2b	The researchers will conduct an experiment to	"Conduct" as a verb	56	3.47	16.1

	evaluate the proposed method.				
E3	The rhyth m of the orchestra changed unexpectedly.	Unexpected space inside a word	37	2.43	15.2
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Time, date, numerals and an unfamiliar word	81	5.56	14.6
T1	Telugu sentence about neuropharmacological research and the effect of new drugs.	Long loan word, rare conjuncts and vowels	66	4.50	14.7
T2a	“His grip on the rope is very strong” (□□□□□ meaning grip).	Context-dependent Telugu word	31	2.02	15.3
T2b	“She came to the ceremony wearing a silk saree” (□□□□□ meaning silk).	Context-dependent Telugu word	37	2.08	17.8
T3	“The tiger walked silently through the forest.”	General sentence with geminates	31	1.83	16.9
T4	Telugu meeting announcement containing a year, ordinal, time and punctuation.	Numbers, ordinal, time and comma	120	7.00	17.1

5.1 Listening observations

ID	Observation
E1	The sentence was clear overall, although slight noise was heard around “antidisestablishmentarianism” and “movement”.
E2a	The generated speech was clean.
E2b	The generated speech was good.
E3	The audio was clear. The system treated the space in “rhyth m” literally and read “rhyth” and “m” as separate items.
E4	The sentence was good overall. “Andaroo” was processed using English letter-to-sound rules, which slightly reduced clarity around the word.
T1	The output was acceptable, with some loss of clarity around rare conjunct and vowel sequences.
T2a	The sentence was very clear.
T2b	The sentence was very clear.
T3	The sentence was good.
T4	This was the most demanding Telugu sentence. Clarity was reduced around the numeral phrases.

Across the ten sentences, the speaking rate ranged from 14.6 to 17.8 phones per second. E4 had the lowest rate and also contained a comma pause of approximately 0.30 seconds. These listening observations come from one listener and one selected speaker per language. They are useful for an initial review, but they do not replace a formal Mean Opinion Score (MOS) or Comparative Mean Opinion Score (CMOS) evaluation.

6. Variance-adaptor analysis

6.1 Measurements

For each input phoneme, the variance adaptor predicts duration, pitch and energy. Duration is measured in mel frames, where one frame is approximately 11.6 ms. Pitch is reported as fundamental frequency (F0) in hertz after reversing the training-set normalisation. Energy is the L2 norm of the STFT frame, following the FastSpeech 2 feature definition.

The inference script saves these predictions as CSV files under Results/<Language>/variance_data/<ID>.csv. Separate duration, pitch and energy plots are stored under Results/<Language>/variance_adaptor_plots/. Figures 3–12 in the original results set combine the three predictions for each sentence on a shared phoneme axis. Pause tokens are shown in grey, and shaded regions identify the parts discussed in the analysis. Since pitch and energy are predicted per phoneme, their plotted values appear as step-like curves.

6.2 Summary of the ten test sentences

ID	Phones	Mean phone	F0 range (Hz)	F0 first → last	Final phone /	Terminal ΔF0	Vowel / obstruent	Ratio	Vowels <20

		(ms)		third	median	(Hz)	energy		
E1	87	59	119–183	143 → 129	2.6×	+5	101 / 29	3.5×	0
E2a	51	61	112–155	137 → 116	3.0×	-2	99 / 21	4.7×	0
E2b	56	62	114–172	142 → 127	1.6×	-12	97 / 22	4.3×	0
E3	37	66	116–180	140 → 132	3.2×	+34	95 / 19	5.0×	0
E4	81	66	104–193	155 → 128	1.0×	-23	93 / 20	4.6×	0
T1	66	68	119–276	217 → 172	1.0×	-28	65 / 36	1.8×	1
T2a	31	65	144–261	210 → 178	1.2×	-34	74 / 13	5.9×	0
T2b	37	56	131–228	173 → 185	1.0×	-17	71 / 22	3.2×	0
T3	31	59	142–256	217 → 184	1.2×	-42	90 / 16	5.5×	0
T4	120	58	131–234	182 → 174	1.0×	+25	54 / 39	1.4×	6

Mean phone duration excludes pause tokens. The first-to-last-third F0 values are duration-weighted means. “Final phone / median” compares the duration of the last phone with the median phone duration in that sentence. Terminal ΔF0 is the final phone's F0 minus the mean F0 in the last third. The energy columns show average predicted energy for vowels and voiceless obstruents. The ratio is vowel energy divided by obstruent energy.

6.3 English test sentences

E1: Long out-of-lexicon word

The 28-phone word “antidisestablishmentarianism” was synthesised without any unusually long phone durations inside the word. Its predicted phone durations ranged from 12 to 128 ms. The longest phone in the complete sentence was the final Z, at 151 ms, or 2.6 times the sentence median. This is consistent with phrase-final lengthening.

Pitch reached 183 Hz on the opening syllables of the long word and then fell to around 119–126 Hz near the primary-stressed EH1 R IY0 sequence. The mean F0 decreased by 14 Hz from the first third to the last third. Vowels had a mean predicted energy of 101, compared with 29 for voiceless obstruents. The 3.5× contrast was the lowest among the English sentences, which is consistent with the slight noise noted during listening.

Figure 3. E1: predicted duration, pitch and energy for each phoneme (source: variance_data/E1.csv).

E2a and E2b: “Conduct” as a noun and a verb

The English front-end produced different stress patterns for the two uses of “conduct”: K AA1 N D AH0 K T for the noun and K AA0 N D AH1 K T for the verb. The variance adaptor reflected this difference. In the noun, the stressed AA1 lasted 58 ms and had predicted energy of 106. In the verb, the unstressed AA0 was reduced to one frame (11.6 ms) with energy 71, while stress shifted to AH1, which lasted 70 ms and had energy 110. The noun's unstressed AH0 lasted 58 ms.

The contrast was therefore visible mainly in vowel duration and energy, rather than pitch. Pitch remained almost flat across the noun (134, 134, 134 and 133 Hz for AA1, N, D and AH0). In the verb, it fell from 149 to 114 Hz. Both sentences also showed phrase-level timing. E2a ended with a 174 ms final S, three times the median phone duration. In E2b, the Z at the end of “researchers” lasted 174 ms and had energy 21, marking a boundary after the subject. The highest pitch in E2b, 172 Hz, occurred on the stressed IY1 in the same word.

Figures 4 and 5. E2a and E2b: predicted duration, pitch and energy for the noun and verb uses of “conduct”.

E3: Unexpected space within a word

The input contains a space in “rhyth m”, and the text front-end preserved it. As a result, “rhyth” was converted to R AY1 DH, while the separate “m” was read as the letter name EH1 M. The two vowel portions of the letter name lasted 70 and 81 ms, and no pause token was inserted between the two items.

The final syllable of “unexpectedly” had the longest phone in the sentence: IY0 lasted 186 ms, or 3.2 times the median. Its F0 rose to 167 Hz from 133 Hz on the preceding L. This produced a terminal rise of 34 Hz relative to the last-third mean, the largest rise in the set and the only rising ending among the English examples.

Figure 6. E3: the isolated “m” token and its predicted variance features.

E4: Time, date, numerals and a comma

The comma in E4 was represented by a pause token lasting 302 ms, with predicted energy of 11. The mean vowel energy in the sentence was 93. The pause followed the final /s/ in “twenty six”, which lasted 174 ms and had energy 12. Together, the final consonant and pause created a low-energy interval of approximately 0.48 seconds.

Pitch remained around 132–133 Hz through the pause, suggesting that the second clause continued from the pitch level of the first rather than starting again at a new level. The expanded forms of the time and date—“ten forty five A M” and “twelve November twenty twenty six”—had ordinary durations. The letters A M were pronounced as EY1 EH1 M.

E4 had the largest overall F0 decline among the English examples, from 155 Hz in the first third to 128 Hz in the last third. The highest value was 193 Hz on “meeting”, while the lowest was 104 Hz on the final phone. The unfamiliar word “andaroo” was converted into six phones (AE0 N D AA1 R OW0) with plausible durations; AA1 lasted 93 ms and OW0 lasted 81 ms.

Figure 7. E4: predicted variance features, including the comma pause.

6.4 Telugu test sentences

T1: Loan word and less common sound sequences

T1 had the widest F0 range in the test set, from 119 to 276 Hz. The sentence began with a rise across the first syllables of the loan word: N at 184 Hz, Y at 222 Hz, UU at 237 Hz, R at 267 Hz and OO at 276 Hz. Pitch then declined, with the duration-weighted mean falling from 217 Hz in the first third to 172 Hz in the last third.

The AU phone in □□□□□ had the longest duration in the sentence, 151 ms. It also had the lowest F0, 119 Hz, and low predicted energy of 26. The vowel-to-obstruent energy ratio was 1.8×, lower than in the Telugu sentences judged clearest. This matches the reduced clarity heard around rare conjunct and vowel sequences.

Figure 8. T1: predicted duration, pitch and energy for the sentence containing the long loan word.

T2a and T2b: □□□□□ meaning “grip” and “silk”

Both sentences contain the Telugu word □□□□□, but the intended meaning changes with the surrounding sentence: “grip” in T2a and “silk” in T2b. The front-end converted the word to the same phone sequence, P A TT TT U, in both cases. The predicted durations were also almost identical. A lasted 46 ms, each TT segment lasted 58 and 46 ms, and U lasted 35 ms in both sentences; P lasted 93 ms in T2a and 81 ms in T2b.

The energy predictions were similar as well: vowel A had energy 83 in both sentences, while U had energy 80 in T2a and 70 in T2b. The pitch shape rose across the geminate cluster towards the final vowel. The overall pitch register was higher in T2a by approximately 27–40 Hz. Both sentences were judged very clear, suggesting that the surrounding words provided enough context for the intended meaning even though the pronunciation sequence was the same.

The wider sentence contours were different. T2a reached 261 Hz on the final vowel of the first word □□□□ and then declined from 210 to 178 Hz across the sentence. T2b was the only sentence in the study with a rise in mean

F0 from the first third to the last third, increasing from 173 to 185 Hz. It began at 131 Hz on the long initial AA in □□□ and reached 228 Hz on the EE in □□□□□□□.

Figures 9 and 10. T2a and T2b: predicted variance features for the two uses of □□□□□.

T3: General sentence with geminates

T3 was the shortest test sentence, with an audio duration of 1.83 seconds. The geminate SH SH in □□□□□□□□□□□ was represented by two consecutive segments lasting 128 and 104 ms, for a combined duration of 232 ms. This is around 2.2–2.5 times the 93–104 ms duration of a single SH in T1 and T4. The result indicates that the model learned the length contrast without a separate explicit duration feature for geminates.

F0 reached 256 Hz on the first word □□□□ and declined to 142 Hz on the final phone. T3 also had the highest vowel-to-obstruent energy contrast among the Telugu sentences, at 5.5×, which is consistent with its good listening result.

Figure 11. T3: the two consecutive SH segments in the geminate sequence.

T4: Year, ordinal, time and punctuation

T4 was the longest input, containing 120 phones and producing 7.00 seconds of audio. The number reader expanded the year, ordinal and time into Telugu words: □□□□□ □□□ □□□□ □□□, □□□□□□□□□ and □□□ □□□□ □□□. The comma was represented by a pause token lasting 81 ms, with energy 6 and F0 of 131 Hz. This was shorter than the 302 ms pause measured in English E4.

The overall F0 changed only slightly, from 182 Hz in the first third to 174 Hz in the last third. There were several phrase-level peaks, with the highest value of 234 Hz on □□□□□□. The sentence ended with a rise to 199 Hz, which was 25 Hz above the last-third mean.

The energy predictions identified three regions with unusually low values: Y A M in □□□□ (14.6, 3.5 and 4.5), BH A in □□□□□□□□□□□□□□□□ (2.3 and 1.1), and A Y AA in □□□□□□□□ (9.9, 2.3 and 5.9). The average vowel energy for the sentence was 54. Six vowels had predicted energy below 20, and five of these were below 10. By comparison, T1 had one such vowel and the other Telugu examples and all English sentences had none. These low-energy regions correspond to the parts where listening clarity was reduced.

Figure 12. T4: low-energy regions and the comma pause.

6.5 Patterns across the test sentences

- Energy and phone type: In all ten sentences, vowels had higher predicted energy than voiceless obstruents. Mean vowel and obstruent energy was 97 and 23 for English (4.3×), and 65 and 29 for Telugu (2.3×).
- Energy and listening observations: The clearest Telugu sentences, T2a, T2b and T3, had vowel-to-obstruent energy ratios of 5.9×, 3.2× and 5.5×. T1 and T4, where clarity was reduced, had ratios of 1.8× and 1.4×. E1 had the lowest ratio among the English examples at 3.5× and was also the sentence with slight noise. This is an observed pattern, not a statistically tested relationship.
- Pitch across an utterance: Mean F0 fell from the first third to the last third in nine of the ten sentences, by 8–45 Hz. T2b was the only exception. The highest F0 occurred on the first content word in six sentences: E1, E2b, E4, T1, T2a and T3. The predicted ranges were 104–193 Hz for English speaker S000 and 119–276 Hz for Telugu speaker TEM.
- Phrase-final lengthening: In four of the five English sentences, the final phone was 1.6–3.2 times the sentence median. E4 was the exception at 1.0×. In Telugu, the final phone was 1.0–1.2 times the median.

- English /r/: All 18 instances of R in the English test sentences were assigned one frame, or 11.6 ms. This may reflect the shortest duration allowed by the MFA alignments, although the alignments were not examined specifically for this behaviour.
- Pause duration: The comma pause was 302 ms in English E4 and 81 ms in Telugu T4. This difference is consistent with the different alignment and feature-extraction procedures used in the two pipelines.

7. Comparison, limitations and possible improvements

The two systems use the same acoustic-model architecture and broadly similar training settings. Differences in the results are therefore likely to be influenced by the datasets, speaker recordings, text front-ends and alignment procedures.

Area	English pipeline	Telugu pipeline
Recording conditions	Speech-recognition-style recordings at 16 kHz; 60 approximated speakers	Studio corpus; two speakers, with TEM clearer than TEF
Text front-end	g2p_en and a pronunciation lexicon; handles common words and heteronyms	Rule-based script-to-phone conversion; covers native Telugu text
Alignment	Pause-aware features	Standard MFA alignments
Training	15,000 steps; about 2.5 hours	22,000 steps
Comma pause in test sentence	0.30 seconds in E4	0.08 seconds in T4
Phrase-final lengthening	1.6–3.2× in four of five sentences	1.0–1.2×
Strong results	Noun/verb stress contrast; numbers, dates and times	Both □□□□□ examples clear; T3 also good
More difficult inputs	Long unfamiliar words, “andaroo” and the stray space in E3	Rare conjuncts and vowels in T1; numeral-heavy T4

The test results point to the text front-end and alignment stages as important factors in speech quality. In English, the main difficulties occurred with unfamiliar words and unusual input formatting. In Telugu, the more difficult cases involved less common sound sequences and number expansion.

A possible common factor is the limited number of examples for rare phoneme sequences in the selected training data. The slight noise in E1 and reduced clarity in T1 both occur around sequences that may be under-represented in approximately five hours of speech. For Telugu, the dictionary contains 68 AU phones and 6 RR phones. In T1, AU also received the longest predicted duration and the lowest F0. This explanation is plausible from the observations, but it was not tested in a separate experiment.

The two □□□□□ examples show that the system can produce a clear pronunciation even when the same phone sequence is used for two meanings. The sequence itself does not encode the difference; the surrounding sentence provides the context. The listening results were clear for both examples.

The energy plots may also be useful when reviewing difficult inputs. T1 and T4 showed low vowel-to-obstruent energy contrast and local drops in vowel energy, alongside reduced clarity during listening. This suggests that per-phoneme energy predictions can help identify regions that may need closer review. It does not establish energy as a reliable quality measure on its own.

Pause timing should be compared carefully across the two languages because the English pipeline used pause-aware alignment while the Telugu pipeline did not. The difference between the 302 ms English pause and 81 ms Telugu pause may therefore reflect preprocessing as well as language-specific prosody.

7.1 Limitations

- Each model was trained on a relatively small amount of speech compared with the original FastSpeech 2 setup.

- The Telugu model uses only two speakers, and the selected speaker recordings differ in clarity.
- The listening assessment involved one listener and one selected speaker per language. No formal MOS or CMOS study was conducted.
- The Telugu pipeline does not yet use the pause-aware feature extraction applied to English.
- The pitch and energy validation losses show a substantial gap from training loss, particularly for Telugu.
- The Telugu number reader and the handling of rare phone sequences need further validation by native speakers and on a larger test set.
- The project uses per-phoneme pitch and energy prediction rather than the frame-level pitch modelling described in the original paper.

7.2 Proposed improvements

1. Apply pause-aware feature extraction to Telugu and continue training. The 81 ms pause in T4, compared with 302 ms in English E4, suggests that silence handling should be checked in the Telugu pipeline as well.
2. Improve generalisation with early stopping based on variance-predictor validation losses, stronger dropout or weight decay, data augmentation, or initialisation from a multilingual pretrained checkpoint.
3. Review the Telugu number reader and pronunciation rules with a native speaker. Adding more number-containing training examples may help with inputs similar to T4.
4. Expand the English pronunciation lexicon with an exception list for foreign and unfamiliar words such as “andaroo”. Decide whether unexpected spaces should be preserved or optionally corrected during normalisation.
5. Carry out a broader evaluation with multiple listeners using MOS or CMOS tests. Pitch and energy ablation experiments would also help determine how much each feature contributes.
6. Bring the variance modelling closer to the original paper by using frame-level pitch and energy and a continuous wavelet transform for pitch. Cleaner, speaker-verified Svarah recordings or a speech-synthesis-focused English corpus would also be useful.

8. Reproducibility and implementation details

The public GitHub repository contains separate folders for the English and Telugu pipelines. Each folder includes the patched FastSpeech 2 code, configuration files, inference script, training and validation lists, speaker information and feature statistics. An inference-only checkpoint of approximately 71 MB is included; it contains model weights only and remains within GitHub's file-size limit.

The Results directory contains generated audio, duration, pitch and energy plots, mel-spectrogram plots, phoneme sequences, per-phoneme CSV files and training-loss curves for both languages. Full checkpoints, including model and optimiser states, are stored in the project Google Drive folder.

8.1 Runtime environment

The experiments were run in Google Colab. CPU runtimes were used for forced alignment, feature extraction and inference, while model training used a Tesla T4 GPU. The Telugu aligner required approximately 4.9 hours on the Colab CPU runtime. Package versions are listed in requirements.txt in the repository.

The original repository specifies older package versions, including PyTorch 1.7 and NumPy 1.19, which are not compatible with the current Colab runtime. The project therefore includes compatibility changes for the versions used during the experiments.

Package	Version / environment
Python	3.13 in Colab; 3.11 in the MFA environment

Montreal Forced Aligner	3.4.3 (conda-forge, installed through micromamba)
PyTorch	2.11
NumPy / SciPy / scikit-learn	2.4.6 / 1.16.3 / 1.6.1
librosa / soundfile	0.11.0 / 0.14.0
pyworld	0.3.5
matplotlib	3.10.0
g2p_en, num2words, tgt, pypinyin, unidecode, inflect, PyYAML	Current releases at runtime

8.2 Main commands

The following commands summarise the main stages of the workflow. The corpus-specific dictionaries, audio and configuration files must be prepared before running them.

```
# 1. Train the forced aligner for the selected language
mfa train --clean -j 2 --output_directory preprocessed_data/<Dataset>/TextGrid \
    raw_data/<Dataset> <dictionary.txt> <dataset>_acoustic.zip

# 2. Extract mel, pitch, energy and duration features
python preprocess.py config/<Dataset>/preprocess.yaml

# 3. Train the model (add --restore_step N to resume)
python train.py -p config/<Dataset>/preprocess.yaml \
    -m config/<Dataset>/model.yaml \
    -t config/<Dataset>/train.yaml

# 4. Generate English test outputs and analysis plots
python inference.py --lang english --dataset Svarah --restore_step 15000 \
    --speaker S000 --out Results/English

# 5. Generate Telugu test outputs and analysis plots
python inference.py --lang telugu --dataset Telugu --restore_step 22000 \
    --speaker TEM --sentences test_sentences_telugu.json --out Results/Telugu
```

In these commands, <Dataset> refers to Svarah for English and Telugu for the Telugu corpus. The configuration files use the same model hyperparameters; the dataset paths, checkpoint paths and validation intervals differ.

9. Conclusion

This project developed and trained two FastSpeech 2 text-to-speech systems, one for Indian-accented English and one for Telugu. Each model was trained using approximately five hours of selected speech. Corpus-specific Montreal Forced Aligner models provided phoneme durations, and a pretrained HiFi-GAN vocoder was used to generate audio. Both systems produced intelligible speech for the ten assignment test sentences.

The English model handled the noun and verb forms of “conduct” with different stress patterns, and it read the tested times, dates and years correctly. The Telugu model produced clear outputs for both meanings of □□□□□ and represented geminate consonants through longer phone durations. A rule-based Telugu front-end was developed as part of the project to make the FastSpeech 2 implementation work with a language that was not supported by the original repository.

The experiments also highlighted areas that need improvement. The text front-end and forced-alignment output had a noticeable effect on the final speech, particularly for unfamiliar words, rare Telugu sound sequences and pauses. Validation loss alone was not enough to choose the best-sounding checkpoint, so listening to the generated

sentences was also important. The per-phoneme variance plots, especially the energy predictions, helped locate parts of the input where clarity was reduced.

The next steps are to apply pause-aware feature extraction to Telugu, improve regularisation of the variance predictors, review the Telugu number reader and pronunciation rules, and carry out a formal evaluation with multiple listeners. These changes would provide a stronger basis for judging the quality and generalisation of both systems.

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Appendix A. Telugu phone inventory

The Telugu front-end uses the following 47 phones. Vocalic r is represented by the two-phone sequence R U. The shared pause token is sp.

Phone group	Symbols
Vowels (12)	A, AA, I, II, U, UU, E, EE, AI, O, OO, AU
Velar, palatal and retroflex stops and nasals	K, KH, G, GH, NG, C, CH, J, JH, NY, TT, TTH, DD, DDH, NN
Dental and labial stops and nasals	T, TH, D, DH, N, P, PH, B, BH, M
Approximants, fricatives and others	Y, R, RR, L, LL, V, SH, SS, S, H